not, with higher-dimensional datasets we don’t have the luxury of using our intuition.

It is time to define a more formal way to assess the quality of a clustering: it’s finally time to talk about evaluation metrics.

To best develop this discussion, we’ll revive our initial example: customer segmentation for our e-commerce site, based on two features, annual income and average monthly bill on the platform. Figure 12.24 shows a synthetic dataset with a realistic distribution (based on data from a real website). The dataset has been preprocessed, normalizing the points: data massaging is a standard step in data science, it helps making sure that features with larger values have the same weight on the final decision; for instance, in our example, we can reasonably assume that the annual salary is in the range $100K-200K, while the average monthly expenses on the website are somewhere between $500-1000: if that was the case, one feature would have values 3 orders of magnitudes larger than the other, and the dataset would be completely skewed. In other words, if we used Euclidean distance to measure how close two points are, the difference in annual salary would contribute overwhelmingly more than the difference in monthly expenses to the final distance between two customers, making the second feature irrelevant.